



ME494 : B.TECH. PROJECT

**Autonomous Control of Quadcopter–Pendulum System for
Induced Oscillations and Stabilization**

22B4508 Tejas Wandile

Guide: Prof. Vivek Sangwan

December 1, 2025

CONTENTS

I	Introduction	2
II	Objective	2
III	Control Architecture	2
III-A	Hardware Setup and Data Acquisition	2
III-B	Communication	2
III-B1	Wireless Data Transport	3
III-B2	Wired Data Transport via Ethernet	3
III-B3	Latency Validation and Comparison	3
III-C	Master PC to Drone Communication	3
III-C1	Hardware Interface	3
III-C2	Protocol Bridge (MAVROS)	3
IV	UAV-Pendulum Model	4
IV-A	Swing-Up Dynamics	5
IV-A1	Drone Body Dynamics	5
IV-A2	Pendulum Dynamics in Relative Frame	5
IV-B	Locked Phase Dynamics (Rigid Body)	8
IV-B1	Rigid Body Properties	8
IV-B2	Equations of Motion (Locked)	8
IV-B3	Sequential Solution and Integration	9
V	Simulation Setup	9
V-A	Numerical Integration	9
V-B	State Transition Logic	10
V-C	Control Strategy	10
V-C1	Outer Loop: Altitude Control (Global)	10
V-C2	Phase 1 Controller: Energy Shaping (Swing-Up)	10
V-C3	Phase 2 Controller: Stabilization and Feedforward	10
VI	Simulation Results	11
VI-A	Phase 1: Swing-Up Dynamics	11
VI-B	Phase 2: Locked Stabilization	11
VII	Results	12
VII-A	Swing-Up Dynamics	12
VII-B	Locked Phase Stabilization	12
VIII	Future Work	12

I. INTRODUCTION

Unmanned Aerial Vehicles (UAVs), particularly quadcopters, have seen rapid adoption in applications ranging from surveillance and inspection to aerial delivery and construction. One of the most challenging configurations in aerial robotics is the transport of suspended payloads, often modeled as a quadcopter-pendulum system. Unlike rigid attachments, a suspended payload introduces significant nonlinearities and underactuated dynamics to the system, as the swing of the load can destabilize the flight controller and degrade tracking performance.

The ability to precisely control the motion of a suspended load is critical for safety and efficiency. While extensive research has focused towards suppressing these oscillations, work towards controlling this swing as dynamical state to be manipulated is limited. Understanding this can help us gain full authority over the pendulum's dynamics: initiating a swing, maintaining it, and extinguishing it on command.

The core challenge lies in the underactuated nature of the system; the pendulum typically has no direct actuation and must be controlled indirectly through the motion of the quadcopter. This work focuses on the problem of executing a multi-phase maneuver that transitions between inducing instability (oscillations) and enforcing stability (damping and locking).

In this work, we present a comprehensive control framework for a quadcopter-pendulum system equipped with an on-board electromagnetic locker. We propose a three-stage control strategy:

- An excitation controller that pumps energy into the system to induce stable oscillations;
- A precision locking maneuver that engages the electromagnet to secure the pendulum at a target angle;
- An active damping controller that rapidly dissipates residual energy to stabilize the system

Ultimately, the findings of this study are expected to contribute to the development of more versatile aerial manipulation systems, particularly in scenarios where precise, instantaneous physical interaction with environmental objects is critical

II. OBJECTIVE

The primary aim of this work is to design and implement an autonomous control system for a **quadcopter-pendulum system**. This project uses the Robot Operating System (ROS) framework to manage communication and control logic. The key goals of this project are:

- To implement a real-time, closed-loop control architecture where the state of the UAV is tracked by a motion capture system, processed by an off-board PC, and the resulting commands are streamed back to the drone via MAVROS.
- To develop a Python-based node capable of interfacing with the drone's flight controller using the MAVROS package.
- To design and validate a control strategy to **intentionally induce** and sustain stable oscillations of the pendulum, targeting a specific amplitude.
- To integrate an on-board electromagnetic locking mechanism and design a control sub-routine for the UAV to engage this locker with the pendulum at a specified point in its swing.
- To design and implement an **active damping controller** to rapidly eliminate all oscillations upon reaching the required amplitude.

III. CONTROL ARCHITECTURE

The proposed control framework relies on a closed-loop architecture that combines external motion capture for state estimation and offboard processing for control. The system consists of three main components: the UAV, the motion capture interface (cameras + data acquisition PC), and the control station.

A. Hardware Setup and Data Acquisition

The UAV is a custom quadcopter equipped with a flight controller and a set of retro-reflective markers. These markers serve as precise reference points for tracking the state of the system.

The sensing environment consists of six motion capture cameras arranged to cover the flight volume. These cameras capture the spatial configuration of the markers in real-time. The raw video data is aggregated by a dedicated **Data Processing PC**, which runs the motion capture software. This software resolves the marker locations to compute the rigid body state of the drone, providing the 3-Dimensional position (x, y, z) and the attitude ($roll, pitch, yaw$) with extremely high accuracy.

B. Communication

The state estimation data is transmitted via a local network from the Data Processing PC to a central **Master PC**. The Master PC serves as the computational core of the system, responsible for running the ROS-based control algorithms. The transfer of data can be done by two possible methods:

1) *Wireless Data Transport*: In the wireless configuration, both the Data PC and the Master PC connect to a common Local Area Network (LAN) via a Wi-Fi router. To facilitate data transfer between these computational units, the system utilizes the distributed networking capabilities of ROS over a TCP/IP layer.

The network topology is established by configuring the environment variables in the `.bashrc` file of both machines:

- `ROS_MASTER_URI`: Points to the IP address of the Master PC, telling the Data PC where to find the `roscore` service.
- `ROS_IP`: Advertises the specific local IP address of the machine itself to ensure bidirectional visibility.

The Master PC hosts the `roscore` service, maintaining a registry of active nodes and topics. The communication follows a specific sequence:

- **Network Registration**: Each machine exports its network configuration, i.e., the IP address (`ROS_IP`) and the Master location (`ROS_MASTER_URI`), allowing the Master to negotiate connections between distinct machines.
- **Peer-to-Peer Streaming**: Once a subscription is established between the publisher (Data PC) and the subscriber (Master PC), the Master achieves the control. The data packets are then transmitted directly between the ROS nodes.

Although this setup offers flexibility for deployment, it is susceptible to signal interference and variable packet delays (jitter), which can destabilize high-frequency control loops.

2) *Wired Data Transport via Ethernet*: To mitigate the latency issues inherent in wireless transmission, a direct wired connection was established using an Ethernet cable. In this configuration, static IP addresses are assigned to the Ethernet interfaces of both PCs, creating an isolated subnet. This dedicated physical link eliminates interference and collision. The software configuration remains identical to the wireless setup (using `ROS_MASTER_URI`), but the transport layer operates over the high-speed Ethernet interface.

3) *Latency Validation and Comparison*: To empirically select the best communication medium, we developed a custom latency benchmarking tool using ROS nodes.

- **Talker Node**: Publishes a message containing the current system timestamp (t_{sent}) to a topic.
- **Listener Node**: Subscribes to the topic and, upon receiving the message, compares the current time ($t_{received}$) with the payload timestamp.

The latency $\Delta t = t_{received} - t_{sent}$ was logged for both configurations. The results confirmed that the Wired Ethernet configuration provided significantly lower average latency and reduced standard deviation compared to the Wireless setup, making it the preferred choice for real-time pendulum stabilization.

C. Master PC to Drone Communication

While high-level processing occurs on the Master PC, the low-level flight control is executed on the drone's on-board flight controller. The link between these two critical subsystems is established using a dedicated Radio Frequency (RF) telemetry pair.

1) *Hardware Interface*: The physical layer consists of two RF transceiver modules. One module is connected to the Master PC via USB, while the counterpart is interfaced with the drone's flight controller via a telemetry port. This setup creates a transparent serial link, enabling bidirectional communication.

2) *Protocol Bridge (MAVROS)*: To interface the ROS-based control logic with the drone's firmware (PX4), the system utilizes the `mavros` package. Running on the Master PC, this node acts as a protocol bridge:

- **Command Transmission**: It converts ROS setpoint messages (e.g., position or velocity commands) into MAVLink packets, which are transmitted over the RF serial link to the flight controller.
- **Telemetry Reception**: Conversely, it decodes incoming MAVLink streams from the drone (IMU data, battery status, arming state) and publishes them as standard ROS topics for the control algorithms to utilize.

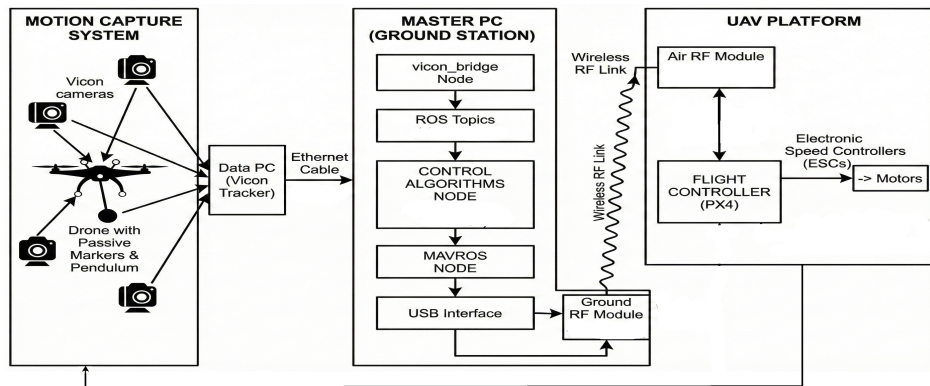


Fig. 1: Control Architecture

IV. UAV-PENDULUM MODEL

The dynamics of the quadcopter-pendulum system are formulated in a two-dimensional (2-D) vertical plane. This planar simplification is justified as the pendulum's motion is primarily restricted to the longitudinal axis during the target maneuvers. Consequently, the control objectives of excitation and stabilization can be achieved by modulating the drone's pitch attitude and total thrust, provided the yaw heading is aligned with the target trajectory prior to initialization. For the mathematical derivation and simulation, the following assumptions are imposed:

- **Coordinate System:** An inertial Cartesian frame $\mathcal{I}$ is defined with the x -axis pointing horizontally to the right and the z -axis pointing vertically upward. Additionally, another non-inertial frame is defined attached to the drone body where z_b is perpendicular to the body and x_b along the body.
- **Sign Convention:** Rotational angles are defined according to the right-hand rule, where counter-clockwise (CCW) rotations are positive.
- **Actuation:** The propulsion system, consisting of front and back rotors, is simplified into two control inputs: a total lift force U_1 and a pitching moment U_2 .
- **Pendulum Mechanics:** The load is modeled as a point mass (bob) connected to the quadcopter's Center of Mass (CoM) via a rigid, massless link.
- **Disturbances:** Aerodynamic drag, joint friction, and cable flexibility are neglected to focus on the core interaction dynamics.

System Configuration and Coordinates

The system kinematics are defined in the 2-D vertical plane using the generalized coordinates $\mathbf{q} = [x, z, \theta, \beta]^T$. The configuration variables and system parameters are defined as follows:

- x, z : The inertial position of the quadcopter's Center of Mass (CoM).
- θ : The pitch angle of the quadcopter with respect to the vertical axis (where $\theta = 0$ implies a hover state).
- β : The angle of the pendulum **relative to the drone's body frame**.
- F_F, F_B : The thrust forces generated by the front and back motors, respectively.
- T : Tension in the pendulum link
- m_f, m_b : The mass of the quadcopter frame and the pendulum bob.
- L : The length of the massless rigid link.
- d : distance of motor from the CoM
- I_y : The mass moment of inertia of the quadcopter about the pitch axis.

The relationship between the actuation forces (F_F, F_B) and the control inputs (Total Thrust U_1 , Pitching Moment U_2) is given by:

$$\begin{aligned} U_1 &= F_F + F_B \\ U_2 &= (F_F - F_B) \cdot d \end{aligned} \quad (1)$$

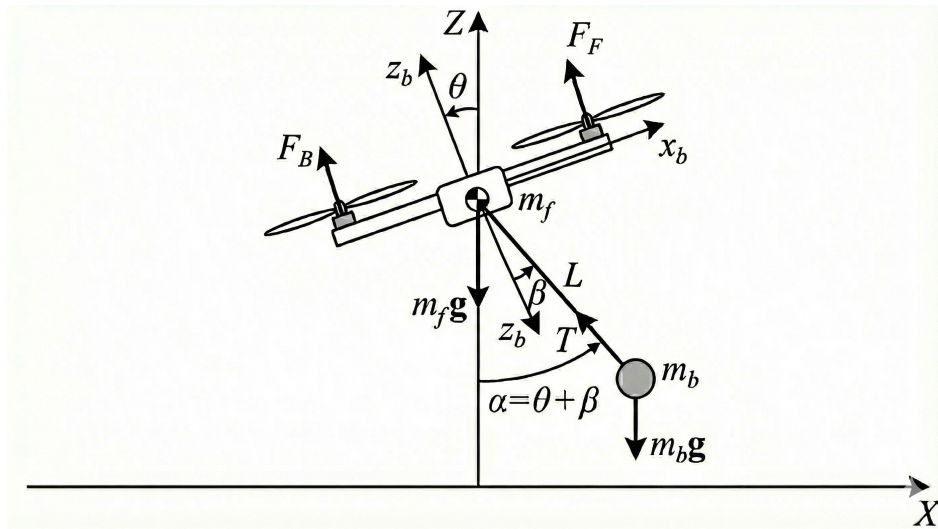


Fig. 2: Drone-Pendulum FBD

A. Swing-Up Dynamics

For the excitation phase, we decouple the system into the quadcopter rigid body dynamics and the relative pendulum dynamics. The quadcopter's motion in the inertial frame is governed by the total thrust U_1 , the gravity vector, and the reaction forces exerted by the pendulum link.

1) *Drone Body Dynamics*: Applying Newton's Second Law to the quadcopter mass m_f :

$$\begin{aligned} m_f \ddot{x} &= -(F_F + F_B) \sin \theta + T \sin(\theta + \beta) \\ m_f \ddot{z} &= (F_F + F_B) \cos \theta - m_f g - T \cos(\theta + \beta) \end{aligned} \quad (2)$$

The rotational dynamics about the pitch axis are governed by the differential thrust (torque) U_2 :

$$I_y \ddot{\theta} = (F_F - F_B) d = U_2 \quad (3)$$

where:

- T is the tension force in the rigid link.
- $U_1 = F_F + F_B$ is the total collective thrust.
- U_2 is the control torque.
- The term $\sin(\theta + \beta)$ represents the horizontal component of the pendulum's pull on the drone.

2) *Pendulum Dynamics in Relative Frame*: To derive the equation of motion for the pendulum, we analyze the dynamics of the bob (mass m_b) in the non-inertial reference frame attached to the accelerating drone.

In the non-inertial drone frame, the bob is subject to real forces (Gravity and Tension) and fictitious (pseudo) forces arising from the acceleration and rotation of the pivot point. The forces acting on the bob are formulated as follows:

1) **Gravity** (F_g): Resolving the global gravitational force into the drone's body frame:

$$F_g = -m_b g \hat{\mathbf{Z}} = -m_b g [\sin(\theta) \hat{\mathbf{i}} + \cos(\theta) \hat{\mathbf{k}}] \quad (4)$$

2) **Tension in the link** (T): Acting along the pendulum rod:

$$T = -T [\sin(\beta) \hat{\mathbf{i}} - \cos(\beta) \hat{\mathbf{k}}] \quad (5)$$

3) **Translational Pseudo Forces** (F_{trans}): Arising from the linear acceleration of the drone ($\ddot{x}, \ddot{z}$):

$$\begin{aligned} F_{\text{trans}} &= -m_b \ddot{x} \hat{\mathbf{X}} - m_b \ddot{z} \hat{\mathbf{Z}} \\ &= -[m_b \ddot{x} \cos(\theta) + m_b \ddot{z} \sin(\theta)] \hat{\mathbf{i}} \\ &\quad + [m_b \ddot{x} \sin(\theta) - m_b \ddot{z} \cos(\theta)] \hat{\mathbf{k}} \end{aligned} \quad (6)$$

4) **Euler Forces** (F_{Euler}): Due to the angular acceleration of the drone frame:

$$F_{\text{Euler}} = m_b L \ddot{\theta} [\cos(\beta) \hat{\mathbf{i}} + \sin(\beta) \hat{\mathbf{k}}] \quad (7)$$

5) **Centrifugal Forces** (F_{Cent}): Due to the angular velocity of the drone frame:

$$F_{\text{Cent}} = m_b L \dot{\theta}^2 [\sin(\beta) \hat{\mathbf{i}} - \cos(\beta) \hat{\mathbf{k}}] \quad (8)$$

6) **Coriolis Forces** (F_{Cor}): Due to the coupling of drone rotation and pendulum swing velocity:

$$F_{\text{Cor}} = -2m_b L \dot{\theta} \dot{\beta} [\sin(\beta) \hat{\mathbf{i}} - \cos(\beta) \hat{\mathbf{k}}] \quad (9)$$

where $\hat{\mathbf{X}}, \hat{\mathbf{Z}}$ represent the global inertial unit vectors, and $\hat{\mathbf{i}}, \hat{\mathbf{k}}$ represent the unit vectors of the drone's body frame.

Summing the individual contributions, the total force $\mathbf{F}_{\text{total}}$ acting on the pendulum bob in the drone's body frame is given by:

$$\mathbf{F}_{\text{total}} = F_i \hat{\mathbf{i}} + F_k \hat{\mathbf{k}} \quad (10)$$

where the component F_i along the lateral body axis is:

$$\begin{aligned} F_i &= -m_b g \sin(\theta) - T \sin(\beta) \\ &\quad - m_b (\ddot{x} \cos \theta + \ddot{z} \sin \theta) \\ &\quad + m_b L \ddot{\theta} \cos(\beta) \\ &\quad + m_b L \dot{\theta}^2 \sin(\beta) - 2m_b L \dot{\theta} \dot{\beta} \sin(\beta) \end{aligned} \quad (11)$$

and the component F_k along the vertical body axis is:

$$\begin{aligned}
F_k = & -m_b g \cos(\theta) + T \cos(\beta) \\
& + m_b(\ddot{x} \sin \theta - \ddot{z} \cos \theta) \\
& + m_b L \ddot{\theta} \sin(\beta) \\
& - m_b L \dot{\theta}^2 \cos(\beta) + 2m_b L \dot{\theta} \dot{\beta} \cos(\beta)
\end{aligned} \tag{12}$$

To eliminate the unknown constraint force (Tension T) and isolate the dynamics of the swing angle β , we resolve the forces along the tangential and radial directions of the pendulum's arc.

We define the radial unit vector $\hat{\mathbf{e}}_r$ (along the link) and the tangential unit vector $\hat{\mathbf{e}}_\tau$ (perpendicular to the link) in terms of the drone's body frame basis ($\hat{\mathbf{i}}, \hat{\mathbf{k}}$):

$$\begin{aligned}
\hat{\mathbf{e}}_r &= -\sin(\beta)\hat{\mathbf{i}} + \cos(\beta)\hat{\mathbf{k}} \\
\hat{\mathbf{e}}_\tau &= \cos(\beta)\hat{\mathbf{i}} + \sin(\beta)\hat{\mathbf{k}}
\end{aligned} \tag{13}$$

a) *Tangential*: Taking the dot product of the total force $\mathbf{F}_{total}$ with the tangential vector $\hat{\mathbf{e}}_\tau$ yields the effective force driving the angular acceleration $\ddot{\beta}$:

$$F_\tau = \mathbf{F}_{total} \cdot \hat{\mathbf{e}}_\tau = F_i \cos(\beta) + F_k \sin(\beta) \tag{14}$$

Substituting the expressions for F_i and F_k :

$$\begin{aligned}
F_\tau = & [-m_b g \sin \theta - T \sin \beta - m_b(\ddot{x} \cos \theta + \ddot{z} \sin \theta) \\
& + m_b L \ddot{\theta} \cos \beta + m_b L \dot{\theta}^2 \sin \beta - 2m_b L \dot{\theta} \dot{\beta} \sin \beta] \cdot \cos \beta \\
& + [-m_b g \cos \theta + T \cos \beta + m_b(\ddot{x} \sin \theta - \ddot{z} \cos \theta) \\
& + m_b L \ddot{\theta} \sin \beta - m_b L \dot{\theta}^2 \cos \beta + 2m_b L \dot{\theta} \dot{\beta} \cos \beta] \cdot \sin \beta
\end{aligned} \tag{15}$$

Simplifying the terms and equating the total tangential force to the tangential inertial term ($m_b L \ddot{\beta}$):

$$m_b L \ddot{\beta} = m_b L \ddot{\theta} - m_b g \sin(\theta + \beta) - m_b \ddot{x} \cos(\theta + \beta) - m_b \ddot{z} \sin(\theta + \beta) \tag{16}$$

Dividing by $m_b L$, we obtain the final equation of motion for the pendulum:

$$\ddot{\beta} = \ddot{\theta} - \frac{1}{L} [g \sin(\theta + \beta) + \ddot{x} \cos(\theta + \beta) + \ddot{z} \sin(\theta + \beta)] \tag{17}$$

b) *Radial*: To determine the tension T , we consider the radial equilibrium of the bob. The tension must counteract the radial components of gravity and the inertial pseudo-forces to sustain the centripetal acceleration.

The total force projected along the radial unit vector $\hat{\mathbf{e}}_r = -\sin(\beta)\hat{\mathbf{i}} + \cos(\beta)\hat{\mathbf{k}}$ is:

$$\sum F_{rad} = T + \mathbf{F}_g \cdot \hat{\mathbf{e}}_r + \mathbf{F}_{trans} \cdot \hat{\mathbf{e}}_r = m_b L (\dot{\theta} + \dot{\beta})^2 \tag{18}$$

Substituting the components derived earlier:

- **Gravity Radial:** $\mathbf{F}_g \cdot \hat{\mathbf{e}}_r = -m_b g \cos(\theta + \beta)$
- **Translational Pseudo Radial:** Using the body frame components derived above:

$$\begin{aligned}
\mathbf{F}_{trans} \cdot \hat{\mathbf{e}}_r &= [F_{trans,i} \hat{\mathbf{i}} + F_{trans,k} \hat{\mathbf{k}}] \cdot [-\sin \beta \hat{\mathbf{i}} + \cos \beta \hat{\mathbf{k}}] \\
&= -F_{trans,i} \sin \beta + F_{trans,k} \cos \beta
\end{aligned} \tag{19}$$

Substituting $F_{trans,i}$ and $F_{trans,k}$:

$$\begin{aligned}
&= -[-m_b(\ddot{x} \cos \theta + \ddot{z} \sin \theta)] \sin \beta \\
&\quad + [m_b(\ddot{x} \sin \theta - \ddot{z} \cos \theta)] \cos \beta \\
&= m_b \ddot{x} (\cos \theta \sin \beta + \sin \theta \cos \beta) \\
&\quad - m_b \ddot{z} (\cos \theta \cos \beta - \sin \theta \sin \beta) \\
&= m_b \ddot{x} \sin(\theta + \beta) - m_b \ddot{z} \cos(\theta + \beta)
\end{aligned} \tag{20}$$

Rearranging the radial equation of motion to solve for T :

$$T = m_b L (\dot{\theta} + \dot{\beta})^2 + m_b g \cos(\theta + \beta) + m_b [\ddot{z} \cos(\theta + \beta) - \ddot{x} \sin(\theta + \beta)] \tag{21}$$

$$\ddot{x} = \frac{1}{m_f} (-U_1 \sin \theta + T \sin(\theta + \beta)) \quad (22a)$$

$$\ddot{z} = \frac{1}{m_f} (U_1 \cos \theta - m_f g - T \cos(\theta + \beta)) \quad (22b)$$

$$\ddot{\theta} = \frac{U_2}{I_y} \quad (22c)$$

$$\ddot{\beta} = -\ddot{\theta} - \frac{1}{L} [g \sin(\theta + \beta) + \ddot{x} \cos(\theta + \beta) + \ddot{z} \sin(\theta + \beta)] \quad (22d)$$

$$T = m_b L (\dot{\theta} + \dot{\beta})^2 + m_b g \cos(\theta + \beta) + m_b [\ddot{z} \cos(\theta + \beta) - \ddot{x} \sin(\theta + \beta)] \quad (22e)$$

Equations (22a) and (22b) describe the translational acceleration of the drone, Equation (22c) describes the rotational pitch acceleration, and Equation (22d) describes the relative angular acceleration of the suspended load. Equation (22e) gives the tension in the link.

The system of equations derived in (22) is coupled: the translational accelerations $\ddot{x}$ and $\ddot{z}$ depend on the tension T , which in turn depends on $\ddot{x}$ and $\ddot{z}$. To enable numerical simulation, we must eliminate the constraint force T and solve explicitly for the accelerations.

Substituting Equation (22e) into the translational equations of motion (22a) and (22b), and grouping the acceleration terms $\ddot{x}$ and $\ddot{z}$ on the left-hand side, we obtain a linear system of the form:

$$\mathbf{M} \begin{bmatrix} \ddot{x} \\ \ddot{z} \end{bmatrix} = \mathbf{C} \quad (23)$$

where $\mathbf{M} \in \mathbb{R}^{2 \times 2}$ is the mass matrix and $\mathbf{C} \in \mathbb{R}^2$ is the force vector.

Let $\gamma = \theta + \beta$. The elements of the mass matrix $\mathbf{M}$ are derived as:

$$\mathbf{M} = \begin{bmatrix} m_f + m_b \sin^2(\gamma) & -m_b \sin(\gamma) \cos(\gamma) \\ -m_b \sin(\gamma) \cos(\gamma) & m_f + m_b \cos^2(\gamma) \end{bmatrix} \quad (24)$$

The force vector $\mathbf{C}$, which contains the Coriolis, centrifugal, gravitational, and thrust terms, is defined as:

$$\mathbf{C} = \begin{bmatrix} -U_1 \sin \theta + m_b g \sin(\gamma) \cos(\gamma) + m_b L (\dot{\beta} + \dot{\theta})^2 \sin(\gamma) \\ U_1 \cos \theta - m_f g - m_b g \cos^2(\gamma) - m_b L (\dot{\beta} + \dot{\theta})^2 \cos(\gamma) \end{bmatrix} \quad (25)$$

By inverting the mass matrix $\mathbf{M}$, we solve for the instantaneous accelerations at each time step:

$$\begin{bmatrix} \ddot{x} \\ \ddot{z} \end{bmatrix} = \mathbf{M}^{-1} \mathbf{C} \quad (26)$$

Once $\ddot{x}$ and $\ddot{z}$ are computed, they are substituted back into Equation (22d) to solve for $\ddot{\beta}$ and Equation (22e) to compute the tension T . Note that the determinant of $\mathbf{M}$ must be checked to ensure non-singularity, which holds true for physical mass parameters ($m_f > 0$).

Once the translational accelerations $\ddot{x}$ and $\ddot{z}$ are obtained from the decoupled linear system, they are substituted back into Equation (22d) to solve for the pendulum's angular acceleration $\ddot{\beta}$:

$$\ddot{\beta} = -\ddot{\theta} - \frac{1}{L} [g \sin(\theta + \beta) + \ddot{x} \cos(\theta + \beta) + \ddot{z} \sin(\theta + \beta)] \quad (27)$$

Additionally, the tension T is computed for analysis purposes using the resolved kinematic variables, confirming the radial force balance on the pendulum bob:

$$T = m_b [L(\dot{\beta} - \dot{\theta})^2 - \ddot{x} \sin(\theta + \beta) + (\ddot{z} + g) \cos(\theta + \beta)] \quad (28)$$

Finally, the simulation advances the system state in discrete time steps using the Forward Euler integration method. For a time step Δt , the generalized coordinates $\mathbf{q} = [x, z, \theta, \beta]^T$ and their time derivatives $\dot{\mathbf{q}}$ are updated as follows:

$$\begin{aligned} \mathbf{q}_{k+1} &= \mathbf{q}_k + \dot{\mathbf{q}}_k \Delta t \\ \dot{\mathbf{q}}_{k+1} &= \dot{\mathbf{q}}_k + \ddot{\mathbf{q}}_k \Delta t \end{aligned} \quad (29)$$

Expanding this for each degree of freedom:

$$\begin{aligned}
x_{k+1} &= x_k + \dot{x}_k \Delta t, & \dot{x}_{k+1} &= \dot{x}_k + \ddot{x} \Delta t \\
z_{k+1} &= z_k + \dot{z}_k \Delta t, & \dot{z}_{k+1} &= \dot{z}_k + \ddot{z} \Delta t \\
\theta_{k+1} &= \theta_k + \dot{\theta}_k \Delta t, & \dot{\theta}_{k+1} &= \dot{\theta}_k + \ddot{\theta} \Delta t \\
\beta_{k+1} &= \beta_k + \dot{\beta}_k \Delta t, & \dot{\beta}_{k+1} &= \dot{\beta}_k + \ddot{\beta} \Delta t
\end{aligned} \tag{30}$$

This recursive update completes one physics cycle, advancing the simulation time $t \rightarrow t + \Delta t$.

B. Locked Phase Dynamics (Rigid Body)

Once the pendulum angle reaches the target threshold ($\beta \geq \beta_{lock}$), the mechanical locking mechanism is engaged. At this instant, the relative degree of freedom β is eliminated ($\beta = \text{const}, \dot{\beta} = 0, \ddot{\beta} = 0$). The system transitions from a multi-body underactuated system to a single, fully actuated rigid body with fixed geometric properties.

To accurately simulate this phase, the dynamics are reformulated with respect to the new combined Center of Mass (CoM) of the drone-pendulum assembly.

1) *Rigid Body Properties*: The total mass of the locked system is simply the sum of the individual masses:

$$M_{sys} = m_f + m_b \tag{31}$$

The position of the new CoM, denoted as (c_x, c_z) in the drone's body frame (relative to the original pivot), is determined by the mass distribution and the locking angle β_{lock} :

$$\begin{aligned}
c_x &= \left(\frac{m_b}{M_{sys}} \right) L \sin(\beta_{lock}) \\
c_z &= - \left(\frac{m_b}{M_{sys}} \right) L \cos(\beta_{lock})
\end{aligned} \tag{32}$$

The effective moment of inertia I_{com} about the new CoM is computed using the Parallel Axis Theorem. Let r_f be the distance from the pivot to the CoM, and r_b be the distance from the pendulum bob to the CoM:

$$\begin{aligned}
r_f^2 &= c_x^2 + c_z^2 \\
r_b^2 &= (L \sin \beta_{lock} - c_x)^2 + (-L \cos \beta_{lock} - c_z)^2
\end{aligned} \tag{33}$$

The total inertia is thus:

$$I_{com} = (I_y + m_f r_f^2) + m_b r_b^2 \tag{34}$$

2) *Equations of Motion (Locked)*: The equations of motion are derived by summing forces and moments about the new CoM. However, because the control inputs U_1 (Thrust) and U_2 (Torque) are generated at the original pivot point (the geometric center of the drone frame), they introduce coupling terms due to the offset CoM.

a) *Rotational Dynamics*: The total moment about the new CoM consists of the aerodynamic control torque U_2 and the moment generated by the thrust force U_1 acting at a lever arm $-c_x$:

$$I_{com} \ddot{\theta} = U_2 - U_1 c_x \tag{35}$$

This equation highlights a critical dynamic shift: due to the non-zero c_x (since $\beta_{lock} \neq 0$), the lift force U_1 now generates a permanent pitching moment disturbance that the controller must counteract.

b) *Translational Dynamics*: Newton's Second Law applies to the acceleration of the CoM ($\mathbf{a}_{com}$). To maintain continuity with the simulation state variables, we express the equations in terms of the pivot acceleration ($\ddot{x}, \ddot{z}$).

The kinematic relationship between the pivot acceleration and the CoM acceleration is:

$$\mathbf{a}_{com} = \mathbf{a}_{pivot} + \boldsymbol{\alpha} \times \mathbf{r}_{com} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}_{com}) \tag{36}$$

where $\boldsymbol{\alpha} = \ddot{\theta} \hat{\mathbf{j}}$ and $\boldsymbol{\omega} = \dot{\theta} \hat{\mathbf{j}}$.

Resolving the forces along the inertial axes yields the following coupled equations, which match the implementation in the simulation step function:

Horizontal (x):

$$M_{sys} \ddot{x} = -U_1 \sin \theta + m_b L \dot{\theta}^2 \sin(\theta + \beta_{lock}) - m_b L \ddot{\theta} \cos(\theta + \beta_{lock}) \tag{37}$$

Vertical (z):

$$M_{sys} \ddot{z} = U_1 \cos \theta - M_{sys} g - m_b L \dot{\theta}^2 \cos(\theta + \beta_{lock}) - m_b L \ddot{\theta} \sin(\theta + \beta_{lock}) \tag{38}$$

These equations account for the inertial effects of the rigid body rotation (centrifugal and tangential acceleration terms derived from the fixed pendulum mass) acting on the drone's translational trajectory.

3) *Sequential Solution and Integration:* To solve for the instantaneous accelerations at each time step, we exploit the structure of the rigid body equations. The solution proceeds sequentially:

a) 1. *Rotational Acceleration:* First, we solve for the angular acceleration $\ddot{\theta}$. Rearranging Equation (35), we obtain:

$$\ddot{\theta} = \frac{1}{I_{com}} (U_2 - U_1 c_x) \quad (39)$$

This value is computed directly from the current state and control inputs.

b) 2. *Translational Acceleration:* With $\ddot{\theta}$ known, we substitute it (along with the current angular velocity $\dot{\theta}$) into the translational force balance equations.

Let the net external forces acting on the system in the inertial frame be denoted as $F_{ext,x}$ and $F_{ext,z}$:

$$\begin{aligned} F_{ext,x} &= -U_1 \sin \theta + m_b L \dot{\theta}^2 \sin(\theta + \beta_{lock}) \\ F_{ext,z} &= U_1 \cos \theta - M_{sys} g - m_b L \dot{\theta}^2 \cos(\theta + \beta_{lock}) \end{aligned} \quad (40)$$

The pivot accelerations are then solved by isolating $\ddot{x}$ and $\ddot{z}$:

$$\begin{aligned} \ddot{x} &= \frac{1}{M_{sys}} (F_{ext,x} - m_b L \ddot{\theta} \cos(\theta + \beta_{lock})) \\ \ddot{z} &= \frac{1}{M_{sys}} (F_{ext,z} - m_b L \ddot{\theta} \sin(\theta + \beta_{lock})) \end{aligned} \quad (41)$$

c) 3. *State Update:* The system state vector for the locked phase is reduced to 6 degrees of freedom: $\mathbf{x}_{locked} = [x, \dot{x}, z, \dot{z}, \theta, \dot{\theta}]^T$. Using the computed accelerations $[\ddot{x}, \ddot{z}, \ddot{\theta}]^T$, the state is propagated using Forward Euler integration with time step Δt :

$$\begin{aligned} \mathbf{v}_{k+1} &= \mathbf{v}_k + \mathbf{a}_k \Delta t \\ \mathbf{p}_{k+1} &= \mathbf{p}_k + \mathbf{v}_k \Delta t \end{aligned} \quad (42)$$

where $\mathbf{p} = [x, z, \theta]^T$ represents position/orientation and $\mathbf{v} = [\dot{x}, \dot{z}, \dot{\theta}]^T$ represents velocities.

V. SIMULATION SETUP

The dynamical system is simulated using the MATLAB environment. The simulation architecture is divided into two distinct engines—DronePendulum (Swing-up Phase) and DronePendulum2 (Locked Phase)—to handle the discontinuity in the degrees of freedom.

TABLE I: List of Simulation Constants and Parameters

Simulation Settings		
Symbol	Value	Description
T_{sim}	20.0 s	Total Simulation Duration
Δt_{phys}	0.002 s	Physics Integration Step (500 Hz)
Δt_{ctrl}	0.02 s	Control Loop Update Step (50 Hz)
Physical System Parameters		
m_f	1.0 kg	Mass of Quadcopter Frame
m_b	0.2 kg	Mass of Pendulum Bob
l	0.5 m	Length of Pendulum Link
I_y	0.01 kg·m ²	Moment of Inertia (Drone Body)
d	0.2 m	Distance from CoM to Motor
g	9.81 m/s ²	Gravitational Acceleration
β_{lock}	75°	Pendulum Locking Angle
Controller Gains		
$K_{p,z}, K_{d,z}$	2.0, 1.5	Altitude (Z) PD Gains
$K_{p,\theta}, K_{d,\theta}$	5.0, 1.0	Attitude (Pitch) PD Gains
K_{swing}	2.0	Swing-Up Gain (Phase 1)

A. Numerical Integration

The equations of motion are integrated using a fixed-step Forward Euler scheme. A high-frequency physics loop runs at 500 Hz ($\Delta t_{phys} = 0.002$ s) to capture the fast rotational dynamics of the drone, while the control loop operates at 50 Hz ($\Delta t_{ctrl} = 0.02$ s) to mimic the update rate of standard onboard flight controllers.

B. State Transition Logic

The simulation initializes in Phase 1 (8-DOF). At each time step, the pendulum angle β is monitored. The transition logic is as follows:

$$\text{If } |\beta(t)| \geq \beta_{lock} \implies \text{Trigger Phase 2} \quad (43)$$

Upon triggering:

- 1) The final kinematic state of Phase 1 $[\mathbf{q}, \dot{\mathbf{q}}]$ is captured.
- 2) The equivalent 6-DOF rigid body state is computed.
- 3) The physics engine switches to the rigid-body solver, and the controller gains are updated to the Phase 2 set.

C. Control Strategy

A hierarchical control architecture is implemented, separating the fast attitude dynamics from the slower translational dynamics. The control laws are adapted based on the system's operational phase.

1) *Outer Loop: Altitude Control (Global)*: The altitude controller is active in both phases. It computes the total thrust U_1 required to track the desired altitude z_{des} . A Proportional-Derivative (PD) law with gravity compensation is used:

$$a_{z,cmd} = K_{p,z}(z_{des} - z) + K_{d,z}(\dot{z}_{des} - \dot{z}) \quad (44)$$

The total thrust U_1 is then calculated by projecting the required vertical force onto the drone's body-z axis:

$$U_1 = \frac{(m_f + m_b)(g + a_{z,cmd})}{\cos \theta} \quad (45)$$

To prevent actuator saturation, U_1 is clamped to a maximum of $2.5 \times$ the hover thrust.

2) *Phase 1 Controller: Energy Shaping (Swing-Up)*: The objective of Phase 1 is to pump energy into the pendulum. We utilize an energy-based control law that modulates the drone's pitch angle θ to excite the pendulum's natural frequency.

The total energy of the pendulum E relative to the pivot is computed as:

$$E(\beta, \dot{\beta}) = \frac{1}{2}m_b L^2 \dot{\beta}^2 + m_b g L (1 - \cos \beta) \quad (46)$$

The desired pitch angle θ_{des} is determined by the energy error:

$$\theta_{des} = \begin{cases} -K_{swing} \cdot \dot{\beta} & \text{if } E < E_{target} \\ 0 & \text{if } E \geq E_{target} \end{cases} \quad (47)$$

This law effectively "kicks" the pendulum in the direction of its motion ($\theta \propto -\dot{\beta}$ creates an acceleration that reinforces the swing).

3) *Phase 2 Controller: Stabilization and Feedforward*: Upon locking, the objective shifts to stabilizing the horizontal position ($x \rightarrow 0$) and the attitude ($\theta \rightarrow 0$) of the now-unbalanced rigid body.

a) *Vertical Position Control*: A PD controller computes the desired acceleration to return the drone to the origin:

$$a_{z,cmd} = K_{p,z}(0 - z) + K_{d,z}(0 - \dot{z}) \quad (48)$$

b) *Inner Loop: Attitude Control with Feedforward*: The inner loop tracks θ_{des} . In Phase 2, standard PID control is insufficient due to the static torque caused by the offset Center of Mass (CoM). We implement a feedforward torque cancellation term.

The required control torque U_2 is:

$$U_2 = \underbrace{I_{locked}(K_{p,\theta}e_\theta + K_{d,\theta}\dot{e}_\theta)}_{\text{Feedback}} + \underbrace{U_1 \cdot c_x}_{\text{Feedforward}} \quad (49)$$

- **Feedback**: Corrects for transient errors using the updated inertia I_{locked} .
- **Feedforward**: Actively cancels the pitching moment induced by the lift force U_1 acting at the lever arm c_x . This term is critical for preventing the "flipping" instability observed in naive control implementations.

VI. SIMULATION RESULTS

The simulation results, presented in Figure 3, demonstrate the successful execution of the swing-up maneuver and the subsequent stabilization of the locked rigid body. The simulation runs for $T = 20$ s, with the locking event occurring at $t_{lock} \approx 8.6$ s (indicated by the vertical dotted line).

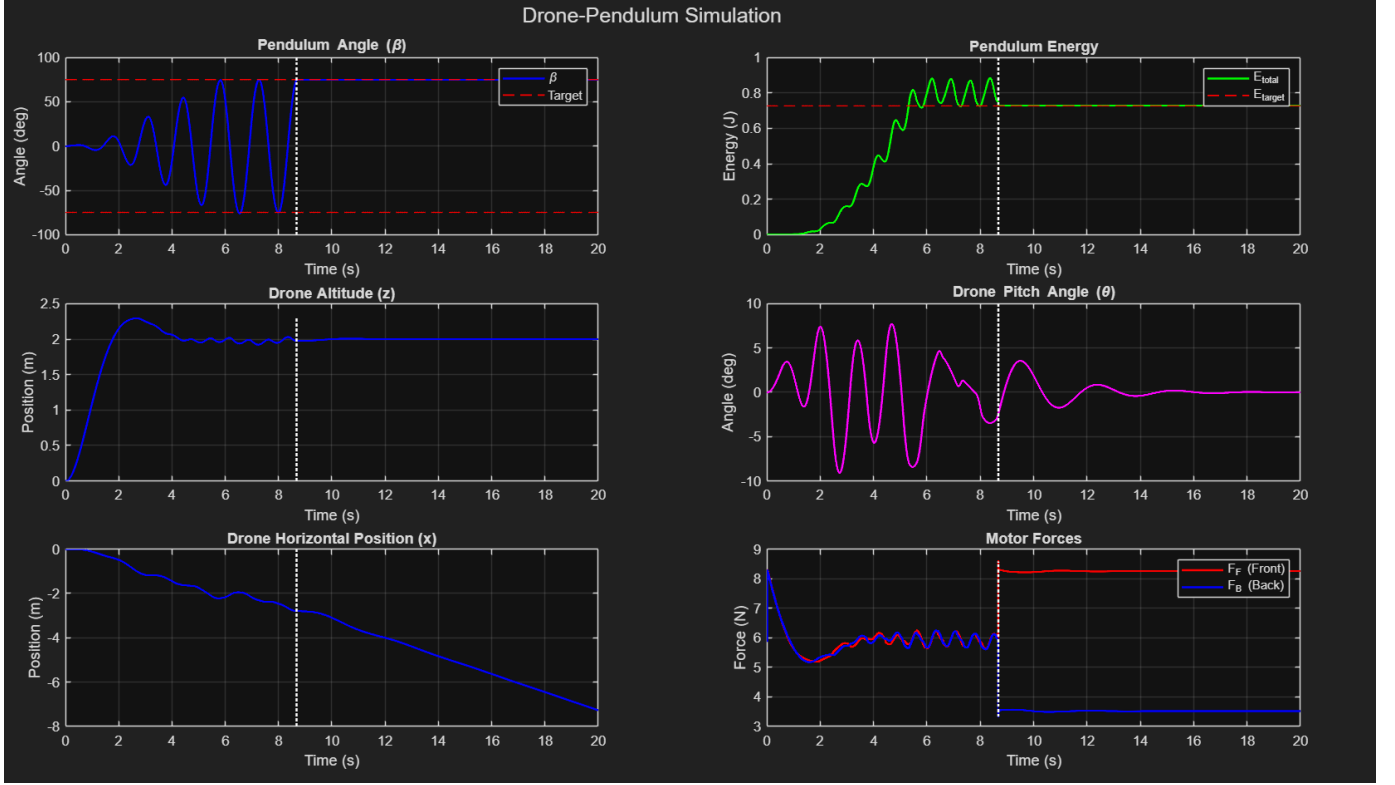


Fig. 3: System response. The vertical dotted line marks the transition from Phase 1 (Swing-Up) to Phase 2 (Locked Stabilization).

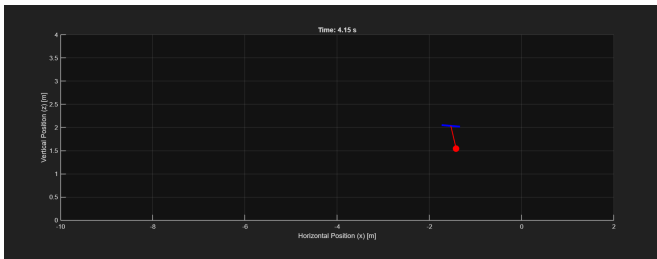
A. Phase 1: Swing-Up Dynamics

The *Pendulum Angle* (β) and *Pendulum Energy* plots illustrate the efficacy of the energy-shaping controller.

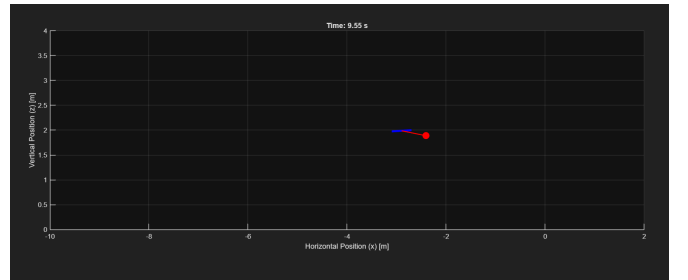
- **Excitation:** The controller successfully pumps energy into the system, increasing the oscillation amplitude from rest. The drone's pitch angle θ (Panel 4) oscillates between $\pm 8^\circ$ in counter-phase to the pendulum to maximize energy transfer.
- **Locking:** At $t \approx 8.6$ s, the pendulum energy crosses the threshold, and β exceeds 75° . The locking mechanism engages, fixing β constant and zeroing the relative velocity.

B. Phase 2: Locked Stabilization

The transition to Phase 2 represents a severe discontinuity in the system dynamics, yet the controller maintains stability.



(a) $t = 4.15$ s (before locking)



(b) $t = 9.55$ s (after locking)

Fig. 4: Drone–pendulum configuration during Phase 2. The quadcopter compensates for the shifted center of mass and prevents flipping while the pendulum remains locked.

- **Attitude Recovery:** The *Drone Pitch Angle* plot shows a transient response immediately after locking. The drone quickly stabilizes to $\theta \approx 0^\circ$. This confirms that the updated inertia tensor I_{locked} and feedforward logic are correctly compensating for the rigid body dynamics.
- **Motor Load Distribution:** The *Motor Forces* plot provides the most critical validation of the control strategy.
 - During Phase 1 ($t < 8.6s$), the front (F_F) and back (F_B) motor forces oscillate symmetrically around the hover value.
 - Upon locking ($t > 8.6s$), a distinct divergence occurs: F_F jumps to ≈ 8.2 N while F_B drops to ≈ 3.5 N.

This permanent offset is the result of the feedforward term $U_{ff} = U_1 \cdot c_x$. The controller automatically biases the front motor to carry the weight of the forward-projecting pendulum, preventing the drone from flipping.
- **Trajectory:** The *Drone Altitude* is held firmly at $z = 2.0$ m. The *Horizontal Position* (x) shows a steady drift after locking. This is the expected behaviour, as the controller is for ($\theta_{des} = 0$), which does not reduce the existing velocity in the x direction, causing the pendulum to move with constant velocity.

VII. RESULTS

The performance of the proposed control strategy was evaluated using a physics-based simulation environment. The results demonstrate successful excitation of the pendulum and subsequent stabilization after the locking event.

A. Swing-Up Dynamics

Figure 3 shows the pendulum angle and its corresponding energy trajectory. The controller efficiently injects energy into the pendulum, increasing the oscillation amplitude from rest. The quadcopter pitch angle oscillates in counter-phase to facilitate energy transfer. When the angle exceeds the locking threshold, the electromagnetic lock is triggered, fixing the pendulum with zero relative velocity.

B. Locked Phase Stabilization

After the transition, the quadcopter-pendulum assembly behaves as a single rigid body. A feedforward-compensated attitude controller ensures stable hovering despite the off-center mass induced by the locked pendulum. The pitch angle converges to zero with only a short transient response. Motor thrust asymmetry observed post-locking confirms proper compensation for the shifted center of mass.

Horizontal motion exhibits slight drift due to non-zero pre-lock velocities, while the altitude remains tightly regulated to the commanded value.

VIII. FUTURE WORK

To continue improving the quadcopter-pendulum framework, several potential developments are identified below.

- **Stabilization in the Horizontal Plane:** Develop a control strategy that ensures stable convergence to a desired horizontal position (x - z plane), eliminating drift following the locking event.
- **Extension to Full 3D Dynamics:** Expand the dynamic model to incorporate yaw and lateral motion, enabling complete 6-DOF control during all operational phases.
- Integrate onboard sensing and real-time feedback to validate the control strategy on an actual drone and suspended payload system.
- **ROS-Based Flight Controller Deployment:** Develop and optimize ROS nodes for implementing swing-up, locking, and stabilization controllers on the physical UAV platform.
- **Locking Mechanism Hardware Refinement:** Improve the electromagnetic latch design for faster response, reliability, and reduced effect on mass distribution.
- **Disturbance Rejection and Safety:** Incorporate robust control methods to account for aerodynamic disturbances, cable flexibility, and actuator saturation limits.